

Enterprise Content Management System (ECMS):

Requirement analysis

1. Entities and Attributes

1. User:

- Attributes: userId, username, email, password, roleId, status, dateCreated, lastLogin
- Relationships: Linked to Role, Permission, Task

2. Document:

- Attributes: documentId, title, content, folderId, categoryId, userId, fileFormatId, versionId, metadataId, dateCreated, lastModified
- Relationships: Linked to Folder, Category, Version, User, FileFormat, Metadata

3. Folder:

- Attributes: folderId, folderName, parentFolderId, userId, dateCreated, lastModified
- Relationships: Linked to User, Document

4. Category:

- Attributes: categoryId, categoryName, description
- Relationships: Linked to Document

5. Permission:

- Attributes: permissionId, roleId, documentId, canView, canEdit, canDelete, canDownload, canShare
- Relationships: Linked to Role, Document

6. Workflow:

- Attributes: workflowId, name, description, status, createdBy, dateCreated, dateModified
- Relationships: Linked to Task, User, Document

7. Version:

- Attributes: versionId, documentId, versionNumber, createdBy, dateCreated
- Relationships: Linked to Document

8. Notification:

- Attributes: notificationId, userId, documentId, message, type, dateSent, status
- Relationships: Linked to User, Document

9. AuditTrail:

- Attributes: auditId, userId, documentId, action, dateOfAction, ipAddress
- Relationships: Linked to User, Document

10. Role:

- Attributes: roleId, roleName, description

- Relationships: Linked to User, Permission

11. **Comment:**

- Attributes: commentId, documentId, userId, content, dateCreated
- Relationships: Linked to User, Document

12. **FileFormat:**

- Attributes: fileFormatId, formatName, extension
- Relationships: Linked to Document

13. **Metadata:**

- Attributes: metadataId, documentId, attributeName, attributeValue
- Relationships: Linked to Document

14. **Task:**

- Attributes: taskId, workflowId, name, status, assignedTo, dateCreated, dateDue
- Relationships: Linked to Workflow, User

15. **Report:**

- Attributes: reportId, name, content, createdBy, dateCreated, lastModified
- Relationships: Linked to User

2. Modules

1. **Document Management:**

- **Functions:** Upload, categorize, search, and manage documents.
- **Features:**
 - Users can upload documents into specific folders.
 - Documents can be categorized for better organization.
 - Full-text search functionality for documents based on metadata or content.
 - Document deletion or restoration based on permissions.

2. **Version Control:**

- **Functions:** Maintain document history, allow versioning, and rollbacks.
- **Features:**
 - Automatic versioning whenever changes are made to documents.
 - Users can view previous versions and roll back if necessary.
 - Maintain detailed logs for changes in each version.

3. **Permission Management:**

- **Functions:** Assign roles and control document access.
- **Features:**

- Role-based access control (RBAC) where permissions (view, edit, delete, download, share) are granted based on the user's role.
- Administrators can assign specific document-level permissions.

4. Workflow Automation:

- **Functions:** Automate document-related tasks such as approvals and publishing.
- **Features:**
 - Define workflows for document approval (multi-level approvals based on document type).
 - Track document processing and task statuses.
 - Automate publishing of approved documents to different categories or folders.

5. Notification System:

- **Functions:** Alert users for document updates, task assignments, and workflow approvals.
- **Features:**
 - Email or system notifications for document status changes, task assignments, and deadlines.
 - Notifications for document uploads, approval requirements, and workflow progression.

6. Audit Trail:

- **Functions:** Track user actions for compliance and security.
- **Features:**
 - Log every user action on documents (view, edit, delete, download) for security and compliance.
 - Store audit logs for legal or regulatory purposes.

7. Reporting Module:

- **Functions:** Generate reports for document usage, access logs, and workflow/task progress.
- **Features:**
 - Generate custom reports based on user roles, document categories, or activity.
 - Track document lifecycle and user interactions for performance analysis.
 - Workflow and task status reports to monitor project progress.

3. Functional Requirements

- **User Management:**
 - Register, authenticate, and manage user profiles.
 - Role assignment (Admin, Contributor, Viewer, etc.) with different permission levels.
- **Document Upload and Storage:**
 - Upload documents with file metadata (file type, size, category, etc.).
 - Organize documents in folders with hierarchical folder structures.

- Perform document operations (edit, delete, move, categorize, version control).
 - **Version Control:**
 - Maintain a version history for all documents.
 - Provide version comparison and rollback functionality.
 - **Workflow and Task Management:**
 - Design workflows for document reviews and approvals.
 - Automate workflow transitions based on task completion.
 - Notifications for tasks and workflow actions.
 - **Search and Metadata:**
 - Search by document name, category, or metadata.
 - Metadata tagging for advanced search options.
 - **Permissions and Access Control:**
 - Role-based access control (RBAC) for folders and documents.
 - Assign different access levels (view, edit, delete, download) based on roles.
 - **Audit Trail:**
 - Track and log all user actions for each document.
 - Record document access and modifications for compliance.
 - **Notifications:**
 - Send notifications for task assignments, document updates, approvals, and other workflow actions.
 - **Reporting:**
 - Generate reports for documents, tasks, workflows, and user actions.
 - Export reports for compliance and performance monitoring.
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4. Non-Functional Requirements

- **Performance:**
 - The system should handle large volumes of documents and users efficiently.
 - Quick document searches, metadata queries, and version comparisons.
- **Scalability:**
 - The system should scale to support multiple departments, thousands of users, and terabytes of documents.
- **Security:**
 - Ensure secure access with encrypted communication (SSL/TLS).
 - Role-based access control to prevent unauthorized access.
- **Usability:**
 - User-friendly interface for uploading and managing documents.

- Simple workflow and notification system to ease task management.
 - **Compliance:**
 - Support for audit trails for compliance with legal and regulatory requirements.
 - Data retention policies for document storage and deletion.
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This **requirement analysis** provides a foundation for developing the ECMS project using Core Java, Collections, and Hibernate as the ORM tool. Each module is defined with its key functionalities, and entities are structured to work efficiently within the system.

Modules Description for ECMS

1. Document Management

Functionality:

- **Upload:** Users can upload documents in various formats, which are stored securely in the system.
- **Organize:** Users can categorize documents into folders and assign them to specific categories for easy retrieval.
- **Search:** An advanced search feature allows users to find documents based on keywords, file types, creation dates, or tags.
- **Preview:** Users can preview documents before downloading to ensure they are accessing the correct files.

Key Features:

- Bulk upload and categorization of documents.
 - Support for various file formats (e.g., PDF, DOCX, XLSX, images).
 - User-friendly interface for navigating and managing documents.
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2. Version Control

Functionality:

- **Track Versions:** Automatically track and store multiple versions of a document, allowing users to view or revert to previous versions.
- **Version History:** Display a comprehensive version history for each document, showing changes made, timestamps, and the user who made the changes.
- **Rollback:** Users can revert to a previous version of a document if needed.

Key Features:

- Easy navigation through different document versions.
 - Comparison tools to view differences between versions.
 - Notifications for users when a new version is created.
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3. Permission Management

Functionality:

- **Role-Based Access Control:** Define user roles and permissions, determining which users can view, edit, or delete documents.
- **Fine-Grained Permissions:** Set specific permissions for individual documents or folders based on user roles.
- **Audit Permissions:** Track and log permission changes for compliance and security purposes.

Key Features:

- Easy assignment of roles and permissions to users.

- Customizable permissions for individual documents and folders.
 - Notifications when permission settings are changed.
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4. Workflow Automation

Functionality:

- **Define Workflows:** Create and manage workflows for document approval, publishing, and other processes.
- **Automated Tasks:** Assign tasks automatically to users based on document status changes (e.g., when a document is submitted for approval).
- **Approval Processes:** Allow users to review and approve documents, sending notifications to involved parties.

Key Features:

- Visual workflow designer for easy configuration.
 - Integration with existing business processes and systems.
 - Tracking of workflow status and task completion.
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5. Notification System

Functionality:

- **Alerts:** Send notifications to users for various events, such as document updates, workflow approvals, or comments on documents.
- **Customizable Notifications:** Users can customize their notification preferences based on their needs.
- **Activity Feed:** Provide a central activity feed for users to track all notifications and updates related to their documents.

Key Features:

- Real-time notifications via email or in-app alerts.
 - Aggregated notifications for improved user experience.
 - History of notifications sent for reference.
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6. Audit Trail

Functionality:

- **Track Changes:** Log all user actions related to documents, such as uploads, edits, deletions, and access.
- **Access Logs:** Maintain records of which users accessed which documents and when.
- **Compliance Reporting:** Generate reports for compliance audits, showing document history and user interactions.

Key Features:

- Detailed logs with timestamps and user IDs for all actions.

- Searchable audit logs for specific documents or users.
 - Compliance alerts for any unauthorized access or changes.
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7. Reporting Module

Functionality:

- **Generate Reports:** Create comprehensive reports on document access, usage statistics, and task progress.
- **Custom Reports:** Allow users to define custom report parameters based on their specific needs.
- **Export Options:** Provide options to export reports in various formats (e.g., PDF, Excel).

Key Features:

- Predefined report templates for common reporting needs.
 - Graphical representation of data for easier analysis.
 - Scheduled report generation and distribution.
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This detailed module description provides a comprehensive overview of the functionalities and features of each module within the **Enterprise Content Management System (ECMS)**.

1. Document Management Workflow

1. User Authentication

- Users log in to the system with their credentials.

2. Document Upload

- Users navigate to the document upload section.
- Users select files to upload and fill in metadata (title, description, category).
- Upon submission, documents are stored in the database and categorized accordingly.
- System generates a notification confirming the successful upload.

3. Document Organization

- Users can create folders and categorize documents.
- Users drag and drop documents into the created folders.
- Changes are saved in the database, and users receive feedback on successful organization.

4. Document Search

- Users enter search criteria (keywords, file types, creation dates).
 - The system fetches matching documents from the database and displays the results.
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2. Version Control Workflow

1. Document Version Creation

- When a document is edited, the system automatically saves the new version.
- Users can view the version history of a document.
- Users can compare different versions to identify changes.

2. Rollback to Previous Version

- Users select a previous version from the version history.
 - The system prompts confirmation for the rollback.
 - Upon confirmation, the selected version is restored as the current version.
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3. Permission Management Workflow

1. Role Definition

- Admins define roles (e.g., admin, editor, viewer) within the system.
- Each role has specific permissions assigned.

2. User Role Assignment

- Admins assign roles to users upon creation or modification of user accounts.
- Users are notified of their assigned roles and permissions.

3. Document Permission Settings

- Users can set document-level permissions based on roles.
 - Changes are saved, and users are notified of updated permissions.
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4. Workflow Automation

1. Workflow Definition

- Admins design workflows for document approval and publishing.
- Workflows consist of predefined steps and assigned users for each step.

2. Document Submission

- Users submit documents for approval, initiating the workflow.
- The system routes the document to the first approver in the workflow.

3. Approval Process

- The assigned approver receives a notification for the pending approval.
 - Approver reviews the document, adds comments if necessary, and approves or rejects.
 - The system updates the document status and notifies the submitting user of the decision.
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5. Notification System Workflow

1. Notification Setup

- Users set their notification preferences (email, in-app alerts) in their profile.

2. Event Triggering

- Events trigger notifications (e.g., document uploads, comments, approvals).
- The system logs these events and prepares notifications.

3. Notification Delivery

- Notifications are sent out to users based on their preferences.
 - Users can view notifications in their notification feed.
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6. Audit Trail Workflow

1. Action Logging

- Every user action (upload, edit, delete, access) is automatically logged in the audit trail.
- The log captures user ID, document ID, action type, timestamp, and details.

2. Accessing Audit Logs

- Admins and authorized users can access audit logs for compliance checks.
- Logs can be filtered by user, document, or action type.

3. Audit Reporting

- Users generate compliance reports from the audit trail.

- Reports can be exported for external review or storage.
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7. Reporting Module Workflow

1. Report Generation Request

- Users navigate to the reporting section and select a report type (e.g., document access, usage statistics).
- Users define parameters for the report (date range, user, document type).

2. Report Creation

- The system compiles the data based on user-defined parameters.
- Reports are generated in the background, with a progress indicator.

3. Report Delivery

- Once generated, reports are available for viewing or download in various formats (PDF, Excel).
 - Users can schedule regular report generation and delivery via email.
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Overall System Workflow

1. User Interaction

- Users log in to the ECMS and interact with various modules based on their roles and permissions.

2. Data Processing

- All actions (uploads, approvals, comments, etc.) are processed in real-time, with appropriate notifications and logs created.

3. System Feedback

- Users receive immediate feedback on actions (success or failure messages) to ensure smooth operation.

4. Administration and Maintenance

- Admins manage user accounts, roles, and system settings through the admin panel.
- Regular maintenance checks are conducted to ensure system integrity and performance.